



BLOOD BANK AND DONOR LOCATOR APP

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Abstract

- **Blood Bank and Donor Locator App** connects blood donors and recipients through an Android application.
- Uses **Node.js, REST APIs, MySQL, Google Maps, and Firebase Cloud Messaging**.
- **Outcome:** Enables faster identification of suitable nearby blood donors.

Problem Statement & Objectives

- **Problem:** Difficulty in finding compatible donors quickly.
- **Objective 1:** Connect donors and recipients.
- **Objective 2:** Enable nearby donor search.
- **Objective 3:** Provide blood requests and instant notifications.
- **Scope:** Donor registration, requests, search, notifications, and admin management.

Existing vs Proposed System

Feature	Existing System	Proposed System
Donor Information	Scattered / difficult to access	Centralized donor database
Donor Search	Manual or limited search	Blood group + location-based search
Availability	Difficult to verify	Updated donor availability
Communication	Delayed communication	Instant notifications
Limitation / Gap	Time-consuming donor identification	Faster matching and communication

Feasibility Study

- **Technical:** Uses Android Studio, Java, Node.js, REST APIs, MySQL, Google Maps, and Firebase Cloud Messaging.
- **Economic:** Uses commonly available development tools and open-source technologies, reducing development cost.
- **Operational:** Simple donor registration, blood request, search, and notification features support easy user interaction.
- **Schedule:** The system can be developed, tested, and documented within the semester timeline.

Results & Testing

- **CRUD Operations:** Successfully implemented for donor and request data.
- **Database Testing:** Verified storage and retrieval of donor, recipient, request, and donation information.
- **Search Testing:** Tested blood-group and location-based donor search.
- **Notification Testing:** Verified notification delivery for blood requests.

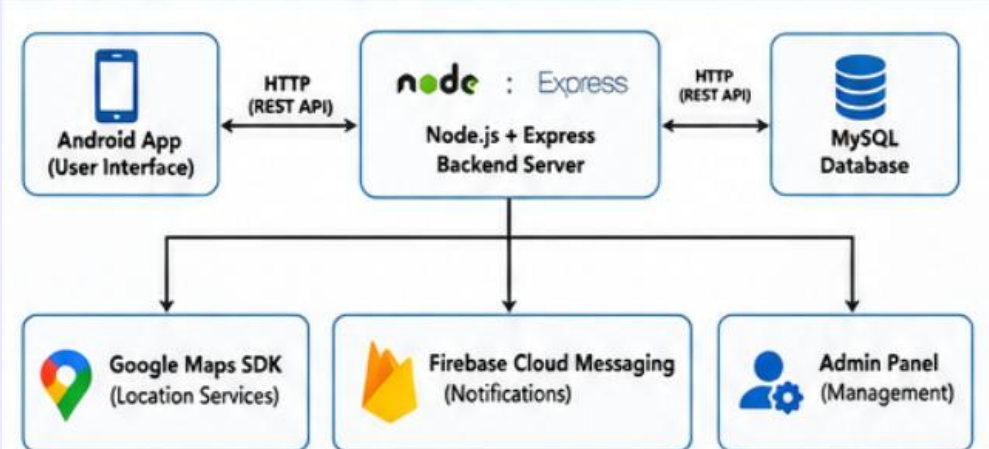
Conclusion

- The proposed system provides a centralized platform for connecting blood donors and recipients.
- It supports donor registration, blood requests, location-based donor search, notifications, donation history, and administrative management.
- The system helps make compatible donor identification and communication more organized and time-efficient.

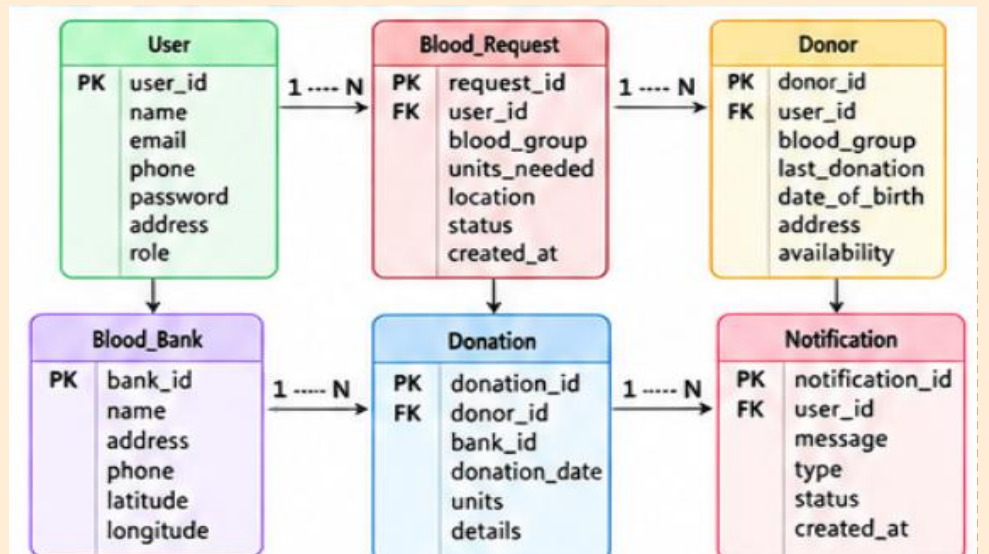
Tools & Technologies

- **Front-End:** React / HTML-CSS-JS
- **Back-End:** Node.js + REST APIs
- **Database:** MySQL
- **Tools:** Git & GitHub, Postman, VS Code, Draw.io

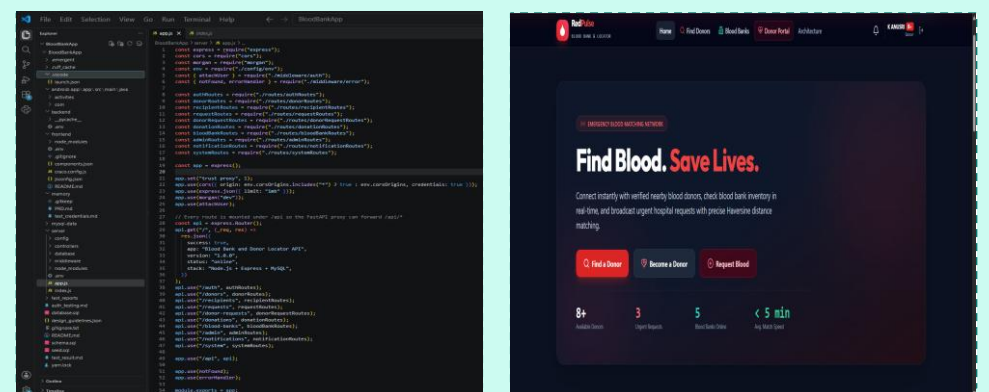
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- AI-based donor matching based on blood group, location, and availability.
- Hospital and blood-bank integration for improved coordination.



Scan to View
GitHub Repository

<https://github.com/kanusri-tmj/Blood-Bank-And-Donor-Locator-App>